

1. How to implement singleton in Java?

Initialization - on-demand holder idiom.

```
public class Singleton {  
    private Singleton () {  
  
        private static class Holder {  
            private static final Singleton Instance =  
                new Singleton();  
        }  
  
        public static Singleton getInstance () {  
            return Holder.Instance;  
        }  
    }  
}
```

2. Features.

- Java 8 Lambdas, Stream API, Optional class, default methods in interface
- Java 11 var for lambda parameters, HttpClient, String methods (isBlank, lines, ^{repeat})
- Java 17 Sealed classes, Pattern matching for instanceof, Records (standardized)
- Java 21 Virtual Threads (Project Loom), Sequenced Collections, Pattern Matching for switch.

3. Prevent memory leak.

memory leaks in Java occur when objects are no longer needed but are still referenced by long-lived objects.

- * Static Collections: Avoid static maps or lists that grow indefinitely; clear them when finished.
- * Listeners / Observers: Always unregister listeners when an object is destroyed.
- * Inner Classes: Non-static inner classes hold a reference to the outer class; use static inner classes where possible.
- * Try-with-resources: Ensure AutoCloseable resources (streams, connections) are closed.

4. Why do we need customized exceptions?

- * Add Meaning: Convey specific business logic errors rather than generic ones.
- * API Clarity: Allow callers to catch specific errors and handle them differently.
- * Encapsulation: Hide internal complexity while providing clear diagnostic information to the user/caller.

5. What's string constant pool for?

The string constant pool is special storage area in the heap. It is used to save memory by ensuring that if two string literals have the same value, they share the same memory address, preventing redundant object creation.

6. Interface vs Abstract Class.

Abstract class : Used when classes ~~to~~ share common state (fields) and behavior. A class can only extend one abstract class.

Interface : Used to define a "contract" or capability. A class can implement multiple interfaces. Since Java 8, interfaces can have default methods, making them even more flexible.

7. How can you avoid race condition?

- * **Immutability** : The best way to avoid race conditions is to use immutable objects.
- * **Synchronization** : Use the synchronized keyword or ReentrantLock to ensure mutual exclusion.
- * **Atomic Variables** : Use java.util.concurrent.atomic classes for lock-free thread safety.
- * **Concurrent Collections** : Use thread-safe collections like ConcurrentHashMap instead of manual synchronization.

8. wait vs Sleep

- * **Thread.sleep()** : A static method that pauses the current thread ~~and~~ for a specific time. It does not release locks held by the ~~str~~ thread.
- * **Object.wait()** : Must be called within a synchronized block. It pauses the thread and releases the lock, allowing other threads to ~~acquire~~ acquire it until notified via notify() or notifyAll().